Project Design Phase
Solution Architecture

Date	03 October 2026
Team ID	SWTID-2026-5297
Project Name	Script-Controlled ACL - Restrict Record Based on Field value
Maximum Marks	4 Marks

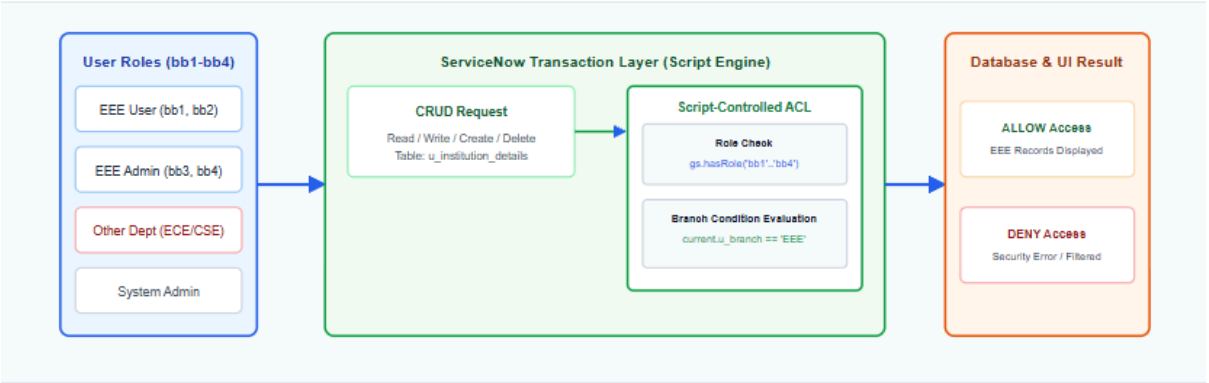
Solution Architecture:

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. Its primary goals are to:

- Find the best tech solution to solve existing cross-departmental data privacy and security vulnerabilities.
- Describe the structure, characteristics, behavior, and security boundaries of the ServiceNow application to project stakeholders.
- Define security features, execution flow, and script-controlled access requirements across custom roles (bb1–bb4).
- Provide technical specifications according to which the security engine is configured, tested, and delivered.

Example -Solution Architecture & Data Flow Diagram:

Figure 1: ServiceNow Script-Controlled ACL Architecture & Security Flow



Architecture Component Specifications:

Component Layer	ServiceNow Module	Security & Architectural Function
User & Identity	Group & Role Management	Assigns custom roles (`bb1` Read, `bb2` Create, `bb3` Write, `bb4` Delete) to targeted departmental users.
Security Logic	Access Control Lists (ACL Script)	Executes server-side JavaScript to validate `gs.hasRole()` and check record context `current.u_branch == 'EEE'` .
Data Storage	Custom Table (`u_institution_details`)	Holds institution records with branch attribute indexing to ensure rapid query execution and isolation.
User Interface	Form & List Views	Dynamically renders filtered list results and toggles visual CRUD elements based on ACL authorization.